

WASP-43b

R-1712

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220209 · created 2026-08-02T15:37:58+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459619.8279 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.013
Transit depth (Rp/Rs)^2	2.31 +/- 0.4 [%]
Orbital Inclination (inc)	82.81 +/- 1.24
Airmass coefficient 1 (a1)	0.8834 +/- 0.0098
Airmass coefficient 2 (a2)	0.089 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#2 - [613, 358]
Optimal Method	PSF photometry
Transit Duration (day)	0.0525 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.152 ± 0.013 vs 0.1594 (deviation 0.57σ)
Duration fit vs published	0.0525 d vs 0.0512 d (deviation 0.37σ)
Mid-time O–C	-2.7 min
Depth SNR	11.6
Residual scatter	1.11 %

Light curve

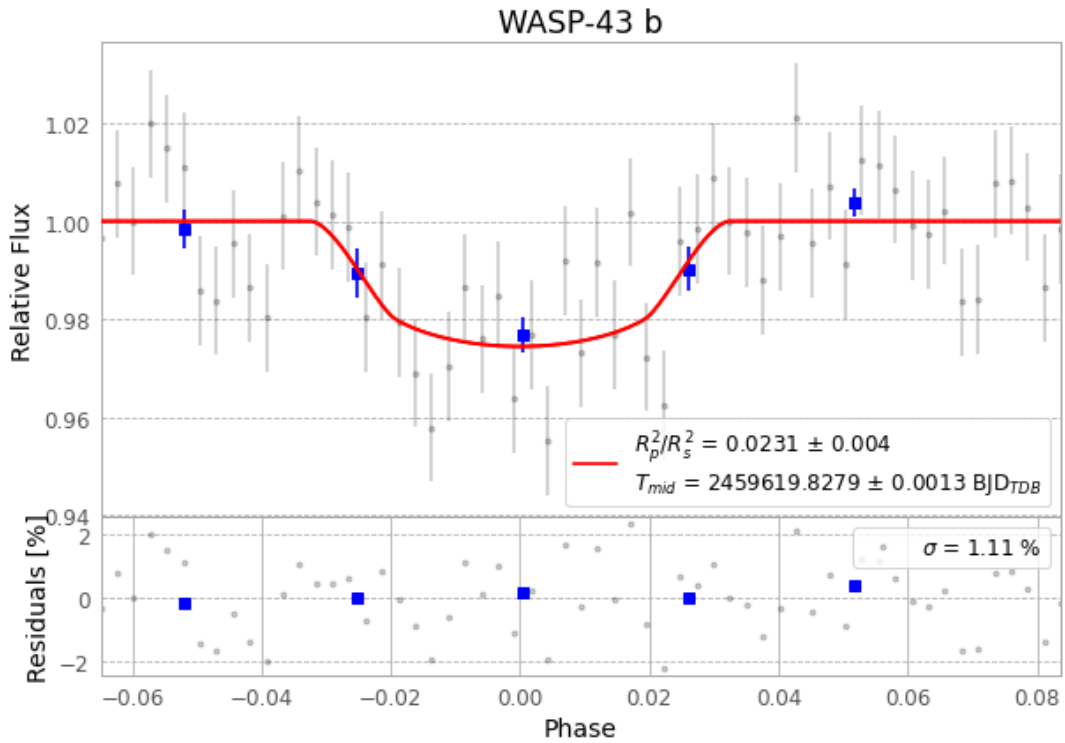


Figure 1 — FinalLightCurve_WASP-43 b_2022-02-08.png (SHA-256 5cdbbb01a0ce95ce...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [304, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2d3165fbb690206f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), WASP-43220209062715.FITS → WASP-43220209092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220209.log (7875276ae498...), FinalLightCurve_WASP-43 b_2022-02-08.png (5cdbbb01a0ce...), FinalLightCurve_WASP-43 b_2022-02-08.csv (29e1f689e30b...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 15:37Z.